



FragMangro: A cross-domain zero-shot model for monitoring fragmented mangrove ecosystems

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Abstract

Monitoring fragmented mangrove ecosystems presents significant challenges due to their sparse distribution and the limitations of traditional detection methods, which often suffer from poor convergence and high rates of false positives and false negatives. To address these issues, we propose FragMangro, a cross-domain zero-shot detection model designed to enhance the accuracy and generalization of mangrove detection. FragMangro integrates parallel multi-scale convolution for progressive dimensional upscaling, improving spectral feature extraction. Additionally, it employs a dynamic learning rate and iterative control algorithm, which adaptively adjusts learning rates and training epochs based on loss monitoring to prevent suboptimal convergence. We conducted extensive experiments in three major mangrove reserves in Guangxi, China. The results demonstrate that FragMangro achieves over 98% accuracy, an average F1 Score exceeding 85%, and the Intersection over Union (IoU) surpassing 74%, significantly outperforming conventional methods such as support vector machines (SVM), random forests (RF), and k-nearest neighbors (KNN). Specifically, in the Beilun estuary, FragMangro exhibits a 0.96% improvement in accuracy over RF, a 17.24% higher F1 Score than SVM, and an 18.08% higher IoU than KNN. Furthermore, the model's adaptive learning rate and dynamic adjustment strategy enhance convergence efficiency and feature representation. FragMangro serves as an effective tool for ecological monitoring and mangrove conservation, with broader implications for environmental protection and sustainable ecosystem management.

Keywords Scattered mangroves · Attention fully convolutional network · Adaptive learning rate and iteration rounds · Deep learning · Zero-shot detection

1 Introduction

Mangroves are unique intertidal wetland ecosystems that provide essential ecological services. They support biodiversity by serving as critical habitats for rare and protected species

and contribute to coastal resilience by mitigating storm surges, preventing erosion, improving water quality, and acting as significant carbon sinks (Saavedra-Hortua et al. 2023; Fu et al. 2023). Given their ecological and environmental importance, effective monitoring of mangrove ecosystems is crucial for biodiversity conservation and climate change mitigation. However, mangroves are increasingly threatened by human activities, biological invasions, and extreme weather events, leading to widespread habitat degradation (Yando et al. 2021; Carugati et al. 2018).

Recent conservation efforts—such as protected area designation and ecological restoration programs—have facilitated partial recovery of mangrove ecosystems (Chen et al. 2009; Wang et al. 2023). Nonetheless, ensuring long-term sustainability requires accurate, timely, and large-scale monitoring to assess ecosystem dynamics and guide data-driven decision-making (Sandilyan and Kathiresan 2012). While traditional field surveys offer detailed observations, they are

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labor-intensive, costly, and impractical for inaccessible or expansive coastal regions (Pham et al. 2019; Thomas et al. 2018). In contrast, remote sensing technologies provide scalable and efficient alternatives by leveraging satellite imagery for systematic ecosystem monitoring (Tian et al. 2017; Kuenzer et al. 2011).

As shown in Fig. 1, mangroves predominantly inhabit intertidal zones, where their muddy terrain poses significant challenges for field-based monitoring. While remote sensing techniques provide a scalable solution for mangrove detection, the effectiveness of deep learning-based approaches depends heavily on high-quality labeled datasets (Dong et al. 2024; Lombard et al. 2023). However, publicly available mangrove datasets often suffer from low annotation accuracy, limiting the generalization capabilities of trained models. Additionally, obtaining labeled mangrove samples through field surveys is costly, time-consuming, and insufficient to meet the demands of data-driven deep learning models. These constraints necessitate new approaches that reduce reliance on extensive labeled datasets while maintaining robust classification performance and generalizability. Most existing mangrove classification approaches rely on conventional convolutional neural networks (CNNs). However, these methods face substantial challenges in detecting small and highly fragmented mangrove patches. The problem is particularly evident in the Guangxi Zhuang Autonomous Region (GZAR), where mangrove forests are sparsely distributed within vast muddy intertidal areas. This geographical characteristic results in a high proportion of mixed pixels and complex spectral variations, complicating sample collection and degrading classification accuracy (Li et al. 2019; Wang et al. 2024; Guo et al. 2021). Moreover, CNN-based models struggle to extract robust features across varying spatial resolutions, often leading to poor convergence and suboptimal classification performance. These limitations highlight the need for alternative strategies that enhance classification accuracy and improve model generalization.

Fig. 1 Mangrove



To address these issues, this study aims to improve the classification accuracy and generalization performance for fragmented and small-scale mangroves in remote sensing imagery. Specifically, we tackle three core challenges: (1) the difficulty of distinguishing small mangrove patches in complex intertidal backgrounds, (2) the high cost and inefficiency of manual labeling in deep learning model training, and (3) the weak cross-domain generalization ability of traditional models for mangroves across different regions.

To this end, we propose FragMangro, a cross-domain zero-shot mangrove detection model designed to enhance classification performance under data-scarce and spatially fragmented conditions. The model leverages Sentinel-2 multispectral satellite data from three major mangrove conservation areas in the GZAR. The main contributions of this study are summarized as follows:

- **FragMangro Model Design:** We propose the FragMangro model, an innovative architecture designed for the extraction of fragmented small patch mangroves. The model enhances feature representation, enabling effective detection of small and fragmented mangrove patches at various spatial resolutions. Extensive experimental results demonstrate that the current network architecture yields the best performance in small patch mangrove extraction tasks.
- **Adaptive Optimization Algorithm (A-LrIr):** We develop a dynamic training mechanism that combines adaptive learning rate scheduling with iterative epoch adjustment. The A-LrIr algorithm addresses the challenge of premature convergence and inefficient training caused by static hyperparameter settings. Compared to conventional optimizers, A-LrIr eliminates the need for manual tuning and improves training efficiency without sacrificing accuracy.
- **Automatic Sample Generation via MREDT:** To overcome the scarcity of labeled samples, we introduce the Mangrove Rough Extraction Decision Tree (MREDT),

which performs coarse yet reliable mangrove region extraction based on spectral indices and confidence thresholds. MREDT automatically generates high-confidence pseudo-labeled samples from remote sensing data, significantly reducing manual annotation efforts and supporting zero-shot model initialization across different domains.

The remainder of this paper is structured as follows: Section 2 reviews existing mangrove detection approaches. Section 3 introduces the study area and data acquisition process. Section 4 details the proposed network architecture and optimization strategies. Section 5 presents the evaluation metrics and baseline comparison models. Section 6 reports experimental results and comparative analyses. Section 7 discusses key findings, and Section 8 concludes with a summary and future research directions.

2 Related work

Current mangrove detection methods can be broadly divided into empirical model-based approaches and techniques utilizing machine learning and deep learning.

2.1 Empirical model-based approaches

Empirical models use threshold segmentation and analysis methods, such as vegetation and water indices, to identify key features in remote sensing. Indices like NDVI and EVI have improved mangrove detection accuracy, with NDVI reflecting vegetation growth and EVI excelling in analyzing high-density vegetation. Xia et al. (2018) introduced the SMRI, integrating texture features and geographic characteristics with an object-based SVM framework to enhance mapping precision. Yang et al. (2022) developed the EMVI using hyperspectral imagery, leveraging green and short-wave infrared bands for better mangrove differentiation, though it is incompatible with Sentinel-2 data and cannot detect underwater mangroves. Chen et al. (2023) addressed this with the Red-Edge Mangrove Index (REMI) based on Sentinel-2 imagery, paired with the Otsu algorithm, achieving 95.68% accuracy and a Kappa coefficient of 0.92. REMI enhances greenness and moisture features, outperforms traditional indices, and eliminates the need for large sample sizes, offering an efficient solution for mangrove detection.

There are limitations to using vegetation indices for mangrove detection. Spectral mixing can blur the distinction between mangroves and other vegetation, while variations in moisture and soil types may alter reflectance, affecting index calculations. Furthermore, low spatial resolution hinders the detection of sparsely distributed mangroves,

and non-linear relationships can lead to unstable correlations between indices and growth conditions. The ecological applicability of different indices also varies, which limits monitoring accuracy. Therefore, it is necessary to integrate other methods to enhance the accuracy of mangrove monitoring.

2.2 Machine learning and deep learning technologies

The advancement of machine learning and deep learning has significantly improved mangrove extraction in remote sensing. Traditional machine learning models provide strong theoretical foundations and high classification accuracy (Huang et al. 2022). For example, Zhang et al. (2021) integrated object-based image analysis with the Random Forest algorithm using Gaofen-2 (GF-2) and Sentinel-2 (S2) imagery. Their findings showed that GF-2 performs better for fragmented mangroves, while S2 excels in identifying submerged mangroves and distinguishing them from terrestrial vegetation. However, supervised learning relies heavily on high-quality annotated data, and unsupervised methods often face challenges in stability and generalization.

Deep learning models, such as Convolutional Neural Networks (CNNs), have further enhanced mangrove extraction by effectively modeling high-dimensional data. CNNs automatically extract features like textures, canopy structures, and contextual relationships, enabling accurate differentiation from other vegetation. Guo et al. (2021) demonstrated CNNs' potential but noted difficulties in classifying samples with similar colors and spatial patterns. Tian et al. (2022) improved U-Net with a channel-attention mechanism to enhance segmentation and reduce noise, though distinguishing spectrally similar plants remained challenging. Sun et al. (2023) optimized U-Net by integrating habitat information during low tide, improving detection of sparse juvenile mangroves and narrow belts but underestimating mangroves in tidal flood-prone areas. Fu et al. (2023) advanced species classification using Adaptive Stacking Ensemble Learning (ASEL) and Multi-Channel Vision Transformer (MPViT) with UAV-derived multispectral imagery, achieving high accuracy in the Beibu Gulf.

Despite these advancements, challenges remain (Kirui et al. 2013; Bunting et al. 2018; Reyes et al. 2024). Deep learning relies on large-scale annotated data, which is limited by cloud cover and spatial resolution constraints, reducing the availability of quality training samples. While effective for high-resolution imagery, models perform poorly on low-resolution data, particularly for small or sparse mangrove patches. Moreover, in complex scenarios with subtle spectral differences caused by human disturbances, model sensitivity and accuracy remain inadequate.

2.3 Zero-shot learning in the mangrove field

Zero-shot learning (ZSL) is a highly challenging task in deep learning, primarily due to the complexity of model parameters and the risk of overfitting when training samples are insufficient. These issues significantly hinder the model's generalization ability. Although deep learning has been extensively applied in remote sensing, research explicitly focusing on zero-shot learning within this domain remains limited (Pradhan et al. 2020; Peña et al. 2024; Lou et al. 2024).

Mangrove ecosystem monitoring presents substantial difficulties due to challenges in sample acquisition, which are driven by geographical inaccessibility, high costs, and ecological sensitivity. These factors make large-scale, high-quality sample collection both costly and complex. In this context, zero-shot learning offers a promising solution. Dong et al. (2024) utilized the zero-shot capabilities of the Segment Anything Model (SAM), combined with the RGRI index, to detect mangroves and salt marshes from UAV imagery, achieving detection accuracies of 83.23% and 95.13%, respectively. However, their method relies heavily on SAM's initial segmentation and manual post-processing, which constrains its full automation potential. Lu and Wang (2021) proposed a large-scale mangrove mapping approach using historical maps to extract training samples, integrating one-class classifiers (SVDD and PUL) with Sentinel-1/2 data. This method achieved a user accuracy of over 97% but still required limited manual correction, which limits its adaptability in previously unstudied regions. Zhao et al. (2024) introduced EIAGA-S, a ground-sample-independent method combining an elite individual adaptive genetic algorithm with the MEVI index. Their approach achieved a mean Intersection over Union (mIoU) of 0.92 and an F1 Score of 0.923 on WorldView-2 imagery. Nevertheless, its performance heavily depends on the design of a robust fitness function, which may lead to misclassifications in areas with sparse mangrove coverage, thus affecting its generalizability.

In the study of mangrove ecosystems, zero-shot learning has been explored to some extent, but overall research in this area remains limited. Given the complex structure and large scale of remote sensing data, enhancing the generalization ability of algorithms presents a significant challenge (Liu 2015; Zhang et al. 2020). Most of the existing research still relies on manual corrections to optimize data processing and model predictions (Zhu et al. 2024; Towle et al. 2024), which not only increases the workload but also has limited effectiveness in improving accuracy. This issue is particularly evident in the extraction of small mangrove patches, where current models perform inadequately and struggle to handle the spatial distribution diversity and scale variations

of mangroves. Therefore, developing more efficient, automated, and adaptable algorithms to address these challenges remains a key research direction in the field of mangrove remote sensing monitoring. This paper addresses the issue of poor performance in extracting small mangrove patches. The proposed method does not rely on manual corrections, as in traditional approaches, to optimize data processing and model predictions. It significantly reduces the need for manual labeling while also demonstrating improved performance in the extraction of small mangrove patches.

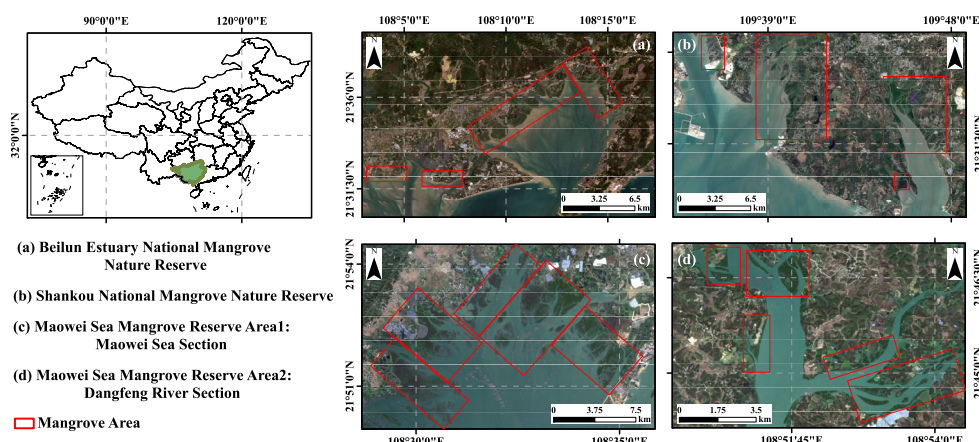
3 Study area and data collection

3.1 Study area

The Guangxi Zhuang Autonomous Region (GZAR), located in southern China near the Beibu Gulf, is home to some of the most extensive mangrove forests in the country. It contains the second-largest mangrove area in China, with forest distributions primarily concentrated along the coastal cities of Beihai, Qinzhou, and Fangchenggang (You et al. 2024). These mangroves serve as critical ecosystems for biodiversity conservation, coastal protection, and carbon sequestration, playing a vital role in promoting environmental sustainability and contributing to climate change mitigation.

This study focuses on three major mangrove reserves within the GZAR, as illustrated in Fig. 2. These include two national-level protected areas—the Guangxi Shankou Mangrove Nature Reserve and the Guangxi Beilun Estuary Mangrove Nature Reserve—as well as one reserve designated at the autonomous region level, the Guangxi Maowei Sea Mangrove Nature Reserve (Yang et al. 2021; Wang et al. 2024). These reserves collectively exemplify the ecological richness of the region and underscore the strategic importance of mangrove conservation. A summary of their key attributes is provided in Table 1.

The spatial distribution and ecological dynamics of mangroves in the Guangxi Zhuang Autonomous Region (GZAR) are analyzed using data from the Sentinel-2 satellite operated by the European Space Agency (ESA) (Data source: <https://browser.dataspace.copernicus.eu/>, accessed on February 29, 2024). Sentinel-2 is an advanced Earth observation platform specifically designed for monitoring terrestrial ecosystems and environmental changes (Van der Meer et al. 2014; Segarra et al. 2020). Among the publicly available satellite datasets, Sentinel-2 provides one of the highest spatial resolutions, making it particularly suitable for ecological monitoring and long-term change detection (Phiri et al. 2020). While commercial satellites such as the WorldView series offer ultra-high-resolution imagery, their applicability

Fig. 2 Map of the study area

is constrained by limited accessibility and high costs. In contrast, Sentinel-2 has been extensively utilized in diverse research domains, including agriculture, forestry, and marine science (Castaldi et al. 2023; Hamrouni et al. 2022; Richardson et al. 2024).

This study employs Sentinel-2 imagery captured on November 21 and 28, 2017, with a spatial resolution of 20 meters. These images constitute the primary dataset for the analysis. The selected scenes are Level-1C top-of-atmosphere reflectance products with less than 10% cloud cover. Following orthorectification and atmospheric correction, the data were converted to Level-2A surface reflectance, enhancing the reliability of surface-level ecological observations. The processed dataset includes nine spectral bands, offering a robust foundation for assessing mangrove distribution patterns and ecological changes. Table 2 summarizes the key parameters of the Sentinel-2 imagery utilized in this study.

4 Methodology

4.1 Mangrove rough extraction decision tree model

This paper introduces the mangrove rough extraction decision tree model (MREDT), initially proposed in the Dynamic Attention Convolutional Network for Mangrove Feature Analysis (DACN-M) framework (Zhao et al. 2024). As illustrated in Fig. 3, MREDT is designed to perform initial rough extraction of mangrove regions.

MREDT enhances the automation and efficiency of mangrove sample identification by integrating multiple spectral indices with key band features. Emphasizing the significance of green, red-edge, and shortwave infrared (SWIR) absorption regions, the model selects bands B3, B8A, B11, and B12 from Sentinel-2 as core inputs. To further enrich feature diversity, bands B2 and B4 are also incorporated. Numerous spectral feature combinations are generated through

Table 1 Overview of mangrove nature reserves in GZAR

Nature Reserve	Location	Area (Hectares)	Mangrove Species	Special Features
Beilun Estuary Mangrove Nature Reserve	Fangchenggang City, Guangxi (108°0'45"E-108°16'6"E, 21°31'43"N-21°37'12"N)	2,680	15 species (10 true mangroves, 5 semi-mangroves), including <i>Aegiceras corniculatum</i>	Largest continuous <i>Aegiceras corniculatum</i> community; only border mangrove area in China
Shankou Mangrove Nature Reserve	Hepu County, Beihai (21°28'22"N-21°37'00"N, 109°37'E-109°47'E)	8,000	12 species, including <i>Rhizophora stylosa</i> , <i>Bruguiera gymnorrhiza</i> , <i>Kandelia candel</i> , <i>Aegiceras corniculatum</i>	Largest and oldest <i>Rhizophora stylosa</i> forest in China; South Asian subtropical mangrove ecosystems
Maowei Sea Mangrove Nature Reserve	Qinzhou City, Guangxi (108°28'35"E-108°54'26"E, 21°44'13"N-21°53'49"N)	2,784	16 species from 11 families, including <i>Bruguiera gymnorrhiza</i> , <i>Kandelia candel</i> , <i>Rhizophora stylosa</i> , <i>Avicennia marina</i> , <i>Aegiceras corniculatum</i>	Largest island group mangrove region; rock-dwelling mangroves; 'Dragon Jing Returns Pearl' landscape

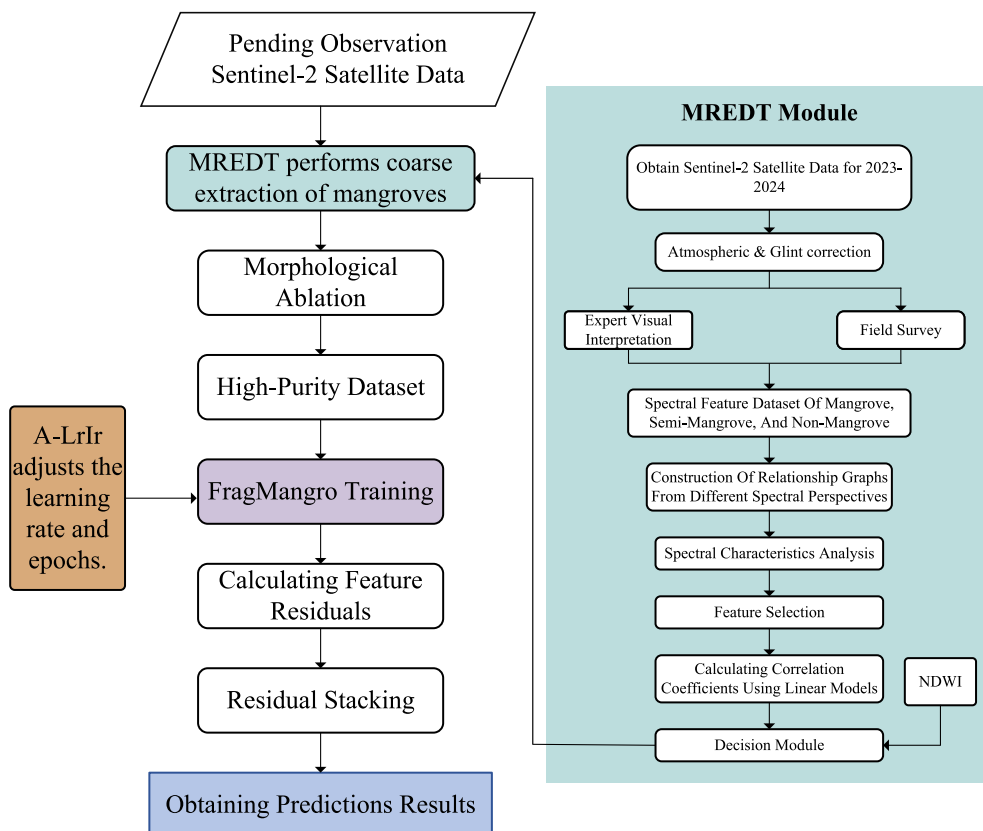
Table 2 Remote sensing image parameter information

Mangrove Nature Reserve	Date	Cloud Cover	Resolution (m)	Band	Wavelength (nm)
Beilun Estuary	21 November 2023	5.64040	20	B2 Blue	490
				B3 Green	560
				B4 Red	665
Maowei Sea	21 November 2023	0.020863		B5 Red Edge	705
				B6 NIR1	740
				B7 NIR2	783
Shankou	28 November 2023	0.602788		B8A NIR4	865
				B11 SWIR1	1610
				B12 SWIR2	2190

arithmetic operations—addition, subtraction, multiplication, and division—resulting in hundreds of candidate features. These features are evaluated using a linear regression model, where the sample-averaged residuals are employed as the loss metric. The optimal feature combination is selected by minimizing the total loss. The resulting MREDT index function is defined as:

$$F(x) = 396.48 \times \left(\frac{\lambda_{865}}{\lambda_{560} + \lambda_{2190}} + \frac{\lambda_{865}}{\lambda_{490} + \lambda_{560}} \right) - 2\lambda_{1610} + 486.28 \quad (1)$$

Here, the central reflectance values at 490nm, 560nm, 865nm, 1610nm, and 2190nm correspond to Sentinel-2 bands B2, B3, B8A, B11, and B12, respectively. The ratio of B8A to B3+B12 effectively captures the contrast in reflectance between high and low wavelength regions in mangrove canopies, while the B8A to B2+B3 ratio enhances the spectral distinction of mangroves in shorter wavelengths. These selected features, together with the observed reflectance gaps in band B11, construct a robust index function $F(x)$ for mangrove identification.

**Fig. 3** FragMangro flowchart

To improve the spatial coherence of the extracted regions, MREDT incorporates the Normalized Difference Water Index (NDWI) to suppress water body interference and applies a morphological erosion algorithm to refine spatial boundaries. This post-processing step reduces edge noise and effectively separates adjacent land cover types, resulting in cleaner and more accurate sample masks.

In conventional remote sensing-based mangrove mapping, generating training data often relies on labor-intensive manual efforts such as digitizing mangrove boundaries or selecting training pixels through expert visual interpretation. This process becomes especially demanding when applied to large or fragmented regions. The proposed MREDT model addresses this issue by automatically identifying high-confidence mangrove regions through rule-based spectral feature analysis and index optimization. These coarse but reliable samples are then used to initialize the FragMangro model training.

4.2 FragMangro model

By integrating the MREDT model with morphological algorithms, this study adaptively refines training samples and introduces a new mangrove detection model, FragMangro. FragMangro is trained using a dynamic algorithm called Adaptive Learning Rate and Iteration Rounds (A-LrIr). This algorithm adjusts the learning rate and the number of iterations throughout training. FragMangro's final predictions are obtained by aggregating residual features extracted during training.

In Fig. 4, the rectangular blocks represent an abstract representation of data features, distinguished by different colors. The blue section corresponds to the feature dimension upgrading module (FDUM), the pink section represents the dimensionality reduction fitting module (DRFM), and the green section indicates the multi-scale convolution module (MSCM) within the feature expansion module. The

dimension labels are located at the top right corner of the blocks. In MSCM, n represents the dimension of the current layer. As shown in Fig. 4, the FragMangro model consists of two core modules: FDUM and DRFM. In order to better extract the spectral features of mangroves in complex environments and improve the extraction accuracy of dispersed small mangrove patches, FragMangro integrates multi-scale features, multiple parallel branch structures, and residual connections. These techniques help FragMangro capture and integrate features from different receptive fields, thereby achieving higher accuracy and better generalization in mangrove detection.

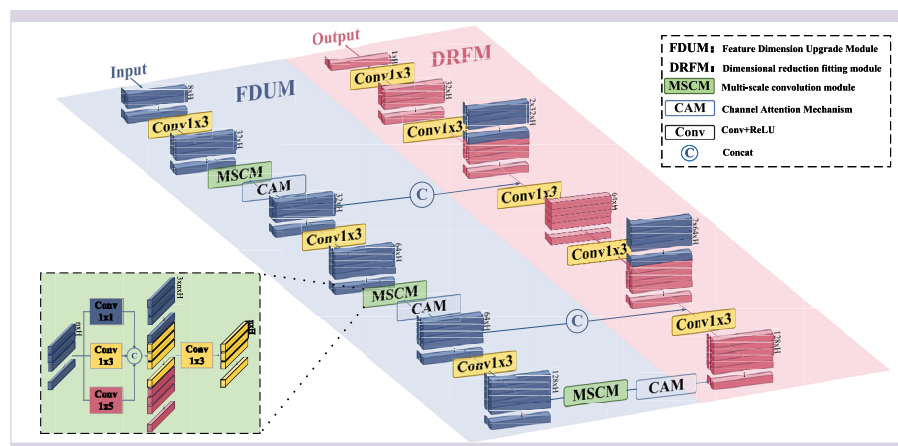
The FDUM module is designed to progressively extract high-level spatial and spectral features from the input images. Its architecture consists of three groups of one-dimensional convolutions with a kernel size of 3, a Multi-Scale Convolution Module (MSCM), and a channel attention mechanism. Through the three groups of one-dimensional convolutions, the dimensionality of the input data is progressively increased to 128. Compared to traditional linear models, this convolution-based architecture offers significant advantages. The convolution operation not only provides a broader receptive field within single-channel data but also enhances the richness of feature representation.

To capture feature information at different scales, the MSCM within the FDUM adopts a parallel multi-branch structure. This design enables more comprehensive feature extraction by processing input data simultaneously at multiple scales. The multi-scale convolution is defined by the following equation:

$$y(x) = \sum_{s \in S} (w_s \times x_s) \quad (2)$$

Where $y(x)$ is the output, x_s is the input at scale s , w_s is the convolution kernel at scale s , S is the set of all scales, and $\times$ represents the convolution operation.

Fig. 4 FragMangro model structure diagram



The MSCM consists of three branches, each using one-dimensional convolutions with kernel sizes of 1, 3, and 5. A 1×1 convolution extracts local features while preserving the spatial dimensions of the input image. A 1×3 convolution balances computational efficiency and receptive field size, making it effective for capturing medium-scale features. A 1×5 convolution improves the model's ability to detect elongated structures, which is useful for identifying strip-like patterns typical of mangroves. The output from each branch is merged using concatenation (cat) to integrate spectral information from different scales. This fusion process enhances feature diversity. MSCM enables the model to capture features at various receptive fields within the same network layer. The fused features are then passed to deeper layers for further extraction and representation. By leveraging multi-scale fusion, the FragMangro model efficiently integrates spectral characteristics at different scales, improving its adaptability to the complex nature of mangroves.

The channel attention mechanism is a critical component of the FDUM module, with its core comprising two parts: a global pooling operation and a multi-layer perceptron (MLP). This mechanism dynamically adjusts the weights of different input channels, enhancing the expression of key features while suppressing less important information. As a result, it improves the model's ability to perceive the importance of features across channels. In FragMangro, the channel attention mechanism is applied after each layer of multi-scale feature fusion, further optimizing the consistency of feature representation. This significantly enhances the model's stability and prediction accuracy in regression tasks.

The DRFM primarily consists of five groups of 1D convolutions with a kernel size of 3, progressively reducing the dimensionality of the features. Ultimately, it outputs the fitting results for the mangrove spectra, allowing FragMangro to better capture critical information in the final stage of feature representation, thereby achieving more precise fitting results. At the same time, DRFM incorporates two residual connections. These residual connections allow lower-level features to be directly introduced into higher-level representations through skip connections. The residual connection is defined by the following formula:

$$\mathbf{X}_{\text{out}} = \mathcal{F}(\mathbf{x}_{\text{in}}) + \mathbf{x}_{\text{in}} \quad (3)$$

Where, x_{in} is the input to the network layer, $F(x_{\text{in}})$ is the function of the network layer skipped by the residual connection, and X_{out} is the output of the residual connection.

In the FragMangro model, residual connection facilitates the fusion of low-dimensional and high-dimensional features, addressing the vanishing gradient problem commonly encountered in deep neural networks while preserving crucial input information. As a result, both the completeness and accuracy of feature representations are improved.

By utilizing residual connections, FragMangro can more effectively harness feature information from multiple levels, leading to enhanced classification performance and improved generalization ability. The dimensionality reduction fitting module gradually reduces the dimensions of feature maps through continuous convolution operations. Combined with a residual connection structure, it ensures the preservation of crucial mangrove features during the reduction process, leading to more effective mangrove recognition.

4.3 Adaptive learning rate and iteration rounds (A-LrIr)

To address potential challenges such as slow convergence or getting trapped in local optima, the FragMangro model integrates a dynamic learning algorithm, referred to as Adaptive Learning Rate and Iteration Rounds (A-LrIr). The pseudocode for the algorithm is presented in Algorithm 1.

A-LrIr incorporates an adaptive learning rate mechanism that dynamically adjusts the learning rate throughout the optimization process, thereby enhancing both convergence efficiency and model stability (as depicted in Fig. 5). In the initial stages of the A-LrIr algorithm, a fixed learning rate is employed to ensure stable convergence during the early phases of training. A-LrIr mitigates the risk of unstable parameter updates that could result from an excessively large learning rate, thus preventing divergence in the early stages of training. As the training progresses, the A-LrIr algorithm continuously monitors the trajectory of the training loss. If the loss demonstrates a decreasing trend, indicating that optimization is proceeding as expected, the algorithm incrementally increases the learning rate to accelerate convergence. On the other hand, if an unexpected increase in loss is detected, the learning rate is reduced to prevent potential divergence caused by over-adjustment of model parameters.

Unlike traditional adaptive learning rate methods, A-LrIr not only incorporates a dynamic adjustment mechanism for the learning rate but also introduces an optimization strategy capable of adaptively regulating the number of training iterations. Rather than simply terminating training upon meeting predefined convergence criteria or reaching a specific performance threshold, the algorithm extends the number of iterations to ensure further optimization of the model beyond its initial convergence. At the same time, when the loss function stabilizes and performance improvements become negligible, A-LrIr can adaptively determine when to automatically terminate the training process. This iterative regulation mechanism fully exploits the model's potential, enhances the final prediction accuracy, and achieves more efficient resource utilization by reducing redundant computations.

A-LrIr effectively balances the trade-off between training speed and model performance by dynamically tuning the learning rate and adaptively extending training rounds.

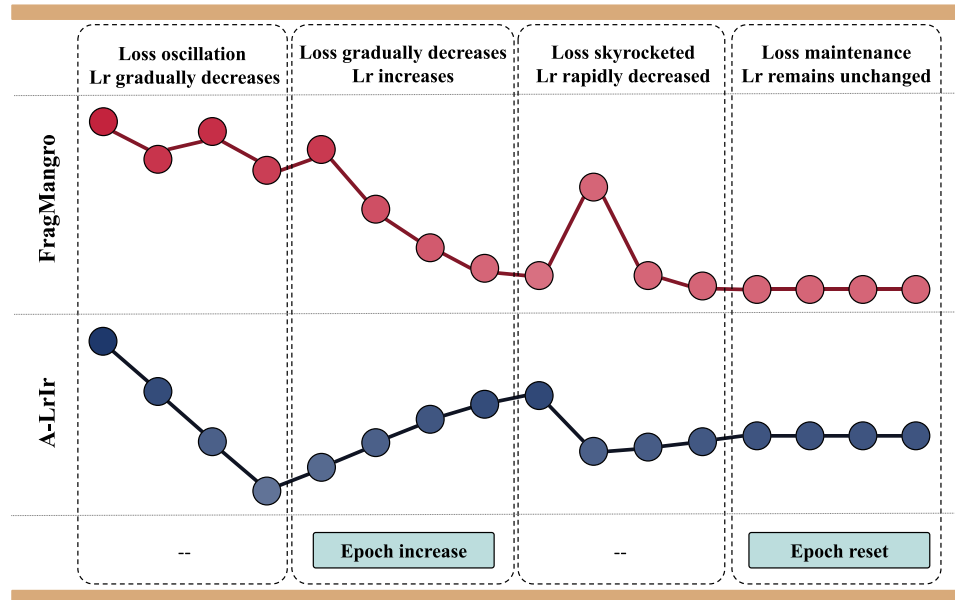
Algorithm 1 A-LrLr.

Input: total training rounds num_epochs , iteration round Y , initial learning rate LR , Loss trend $loss_SG$, loss history $loss_history$, learning rate history $lr_history$, minimum loss $loss_min$;

Output: $epoch$, $loss$, and lr ;

- 1 Initialize $optimizer$ as Adam with model parameters and LR ; $criterion$ as Mean Squared Error Loss;
- 2 Initialize $loss_SG$, $loss_history$, and $lr_history$ with [0];
- 3 $num_epochs \leftarrow 2000$; $epoch \leftarrow 0$; $LR \leftarrow 0.001$; $loss_min \leftarrow None$;
- 4 **while** $epoch < num_epochs$ **do**
- 5 Use criterion to calculate the loss of the model run; Clear the optimizer gradients to zero; Perform backpropagation on the loss; Update optimizer;
- 6 $mid_epochs \leftarrow 100$; $down_epoch_size \leftarrow 250$;
- 7 **if** $(epoch + 1) \bmod mid_epoch = 0$ **then**
- 8 Append current $loss$ to $loss_history$;
- 9 **if** $epoch > mid_epoch$ **then**
- 10 Calculate mid_loss_delete as change in recent losses;
- 11 Update $loss_SG$ based on mid_loss_delete ;
- 12 **end**
- 13 **if** $loss > down_epoch_size$ **then**
- 14 $lr \leftarrow LR$;
- 15 **else**
- 16 $lr \leftarrow LR$ scaled by $loss$ and $down_epoch_size$;
- 17 **end**
- 18 Reinitialize optimizer with new lr ;
- 19 **if** $epoch > 10 \times mid_epoch$ **then**
- 20 Try to calculate $deger_up_judge$ as ratio of recent loss changes;
- 21 **if** $loss_SG$ indicates consistent decrease **then**
- 22 Adjust num_epochs based on $loss$ value;
- 23 **if** $loss < 2$ **then**
- 24 Break loop;
- 25 **end**
- 26 **end**
- 27 **if** $loss_SG$ indicates stable convergence and $LR < 0.01$ **then**
- 28 Increase LR ;
- 29 Print increased LR ;
- 30 **end**
- 31 **if** $loss_SG$ indicates abnormal increase and $LR > 0.0001$ **then**
- 32 Decrease LR ;
- 33 Print decreased LR ;
- 34 **end**
- 35 **if** losses are unchanged **then**
- 36 Break loop;
- 37 **end**
- 38 **if** $deger_up_judge$ indicates abnormal increase **then**
- 39 Adjust LR and $down_epoch_size$;
- 40 Print decreased LR ;
- 41 **end**
- 42 **end**
- 43 Print current $epoch$, $loss$, and lr ;
- 44 **if** $loss_min$ is None or current $loss < loss_min$ **then**
- 45 Update $loss_min$;
- 46 Save model weights to specified path;
- 47 **end**
- 48 **end**
- 49 Increment $epoch$ by 1;
- 50 Append current LR to $lr_history$;
- 51 **end**

Fig. 5 Diagram of A-LrIr



A-LrIr not only accelerates the convergence speed but also significantly reduces the model loss, thus improving the speed and accuracy of mangrove identification.

5 Evaluation methods

5.1 Comparison algorithms

To validate the effectiveness of FragMangro in mangrove detection, we conducted comparative experiments using three traditional algorithms: Support Vector Machine (SVM), Random Forest (RF), and k-Nearest Neighbors (KNN). All algorithms followed the same experimental procedures to ensure comparability of results.

5.1.1 Support vector machine (SVM)

Support Vector Machine (SVM) is a supervised learning algorithm known for its effectiveness in high-dimensional data processing and classification tasks (Hearst et al. 1998). Its strength lies in efficiently handling multidimensional classification and recognition problems. The SVM objective function is defined as:

$$\min_{w,b} \frac{1}{2} \|w\|^2, \text{ s.t., } y_i(w x_i + b) \geq 1 \quad (4)$$

In multispectral remote sensing, each spectral sample is represented as a vector in high-dimensional space, with reflectance values of individual bands as features. SVM uses labelled training samples to construct a model that identifies an optimal hyperplane, maximizing separation

between categories. This results in a robust classifier capable of distinguishing various classes. For mangrove spectral sensing, SVM enables accurate identification and classification of mangroves and other vegetation types. For example, Heumann et al. applied SVM with object-based image analysis and Worldview-2 sensor data to classify mangroves, achieving an overall classification accuracy exceeding 94% (Heumann 2011). This highlights SVM's effectiveness in remote sensing applications.

5.1.2 Random forests (RF)

RF is an ensemble learning algorithm widely used in mangrove classification since its introduction in 2001 (Breiman 2001; Jiang et al. 2021). It combines predictions from multiple decision trees to analyze spectral data. The method involves creating training datasets through bootstrapping, where samples are randomly drawn with replacements from the original dataset. A random subset of features is used for each decision tree. After constructing these trees, predictions are aggregated through majority voting to determine the final classification. The RF model is expressed as:

$$y = \frac{1}{k} [t_1 \cdot \text{prob}(x) + t_2 \cdot \text{prob}(x) + \dots + t_k \cdot \text{prob}(x)] \quad (5)$$

Where y is the predicted probability of each category, t_i is a decision tree, $\text{prob}(x)$ is the prediction of x by the i -th tree, and k is the number of trees in the forest.

The effectiveness of RF in mangrove classification is well-documented. For example, Purwanto et al. compared decision tree and RF algorithms for classifying mangroves in Semilang National Park (Purwanto et al. 2022). Their

findings showed that RF consistently outperformed individual decision trees, providing more accurate assessments of mangrove distribution. This enhanced accuracy is crucial for effective conservation and restoration efforts.

5.1.3 K-nearest neighbors (KNN)

The K-Nearest Neighbors (KNN) algorithm is a widely used non-parametric supervised learning method for classifying samples based on similarity (Guo et al. 2003). KNN assigns a new spectral sample to the category most common among its K nearest neighbors. Distances can be calculated using methods such as Euclidean, Manhattan, or Chebyshev distance, with Euclidean being the most common. The formula for Euclidean distance is:

$$d(x, y) = \sqrt{\sum_i (x_i - y_i)^2} \quad (6)$$

In mangrove spectral recognition, KNN compares spectral similarities without assuming an underlying probability distribution, making it suitable for nonlinear and complex classification problems. In 2016, Silva et al. enhanced KNN for delineating mangrove canopies (Silva et al. 2016). Additionally, Yando et al. used UAV LiDAR data with various algorithms to segment individual trees (Yando et al. 2021), finding that the improved KNN algorithm excelled in accuracy, particularly in low-density areas. This highlights its effectiveness in complex remote sensing applications where data density varies.

5.2 Evaluation indicators

To validate the accuracy of mangrove remote sensing extraction, a confusion matrix was employed to assess the prediction results of mangrove canopy at selected sample points (Deng et al. 2016). In this evaluation process, a 'false negative' refers to pixels that were visually identified as mangrove canopy but were not classified as such in the prediction results. Conversely, a 'false positive' denotes pixels that were incorrectly predicted as mangrove canopy, despite not actually representing it. The classification performance was evaluated using three key metrics: accuracy, precision, and recall.

Additionally, the overlap between actual and predicted mangrove canopies was systematically evaluated. For each actual canopy, the number of intersecting predicted canopies was documented. A high prediction success rate is indicated when the majority of actual canopies overlap accurately with at least one predicted canopy. To rigorously quantify this overlap, the Intersection over Union (IoU) metric was applied, representing the degree of spatial alignment between two regions. IoU is mathematically defined as the ratio of the

intersection area to the union of the actual canopy area and the predicted canopy area.

Precision represents the ratio of the bounding box of the intersecting area to the bounding box of the predicted canopy, while Recall indicates the ratio of the bounding box of the intersecting area to that of the actual canopy. The F1 Score, which synthesizes Precision and Recall, is calculated as their harmonic mean, providing a balanced assessment of detection performance.

6 Experimental setup and results

6.1 The experimental results of the FragMangro model

The FragMangro model was trained using an NVIDIA RTX 4090 GPU in conjunction with the GPU version of PyTorch. The initial learning rate was set to 0.001 and dynamically adjusted through the A-LrIr strategy. The fitting loss of the FragMangro was calculated using the Mean Squared Error (MSE) criterion, while the Adam optimizer was employed to update the learning rate parameters. A-LrIr enables the FragMangro model to achieve optimal convergence during each training session.

Equidistant image sampling techniques were employed to evaluate the effectiveness of the FragMangro model. In the four study areas, validation points were generated by extracting a sample point every six pixels, resulting in a comprehensive validation set. Mangrove areas were delineated through visual interpretation, utilizing high spatial resolution imagery from Google Earth, supplemented by relevant research for attribute annotation.

As shown in Fig. 6, red and white point sets represent mangrove canopies and non-mangrove areas, respectively. equidistant image sampling techniques maintain the original image's class distribution in the validation points while preserving randomness and independence in sampling, providing a robust basis for evaluating the accuracy of different mangrove monitoring models. To comprehensively assess model performance, we compared six algorithms across all study areas: KNN, RF, SVM, FCN, DNN and FragMangro. This comparison not only validated the reliability of the FragMangro model but also highlighted its distinct advantages. Unlike traditional methods like KNN, RF, and SVM, which require labelled samples for training, FragMangro leverages MREDT technology, significantly reducing the data collection burden. For a fair comparison, we extracted 500 positive samples and 1,000 negative samples from the dataset annotated in Fig. 6 to train the KNN, RF, and SVM. FCN and DNN were trained using the training samples extracted by MREDT.

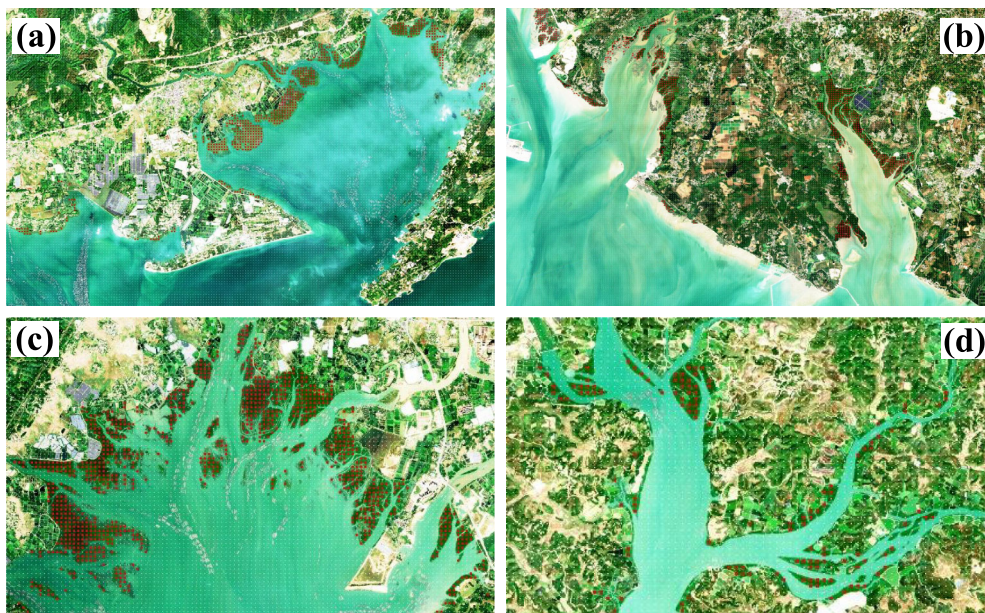


Fig. 6 Mangrove area annotation map (a) Beilun estuary national mangrove nature reserve; (b) Shankou national mangrove nature reserve; (c) and (d) Maowei sea mangrove reserve

Table 3 presents the evaluation metrics for FragMangro, SVM, RF, KNN, FCN, and DNN. Through multi-level feature extraction and the fusion of contextual information, the FragMangro model effectively distinguishes the semantic features of mangrove canopy and non-mangrove areas. In terms of all evaluation metrics, including Accuracy, Precision, Recall, F1 Score, and IoU, FragMangro consistently outperforms the other models. In all study areas, Accuracy remains above 97%, while the F1 Score exceeds 75%. These findings strongly validate the high consistency between the detection results of the FragMangro model and the annotated canopy data, demonstrating that the network architecture of FragMangro can efficiently and accurately extract mangrove areas, including small mangrove patches.

Among the five comparison models, the FCN and DNN models exhibit relatively lower evaluation metrics across four study areas. These two deep learning models are more susceptible to the influence of sample size. A lower proportion of mangrove samples in the training data leads to the models' inability to accurately extract mangrove spectral features when faced with complex scenarios. The FCN model shows significantly higher recall than the other models across the four study areas. However, in the Beilun Estuary and Maowei Sea Reserve Area 2, other metrics are notably lower. This indicates that the FCN model tends to "over-detect" mangroves, successfully identifying the majority of mangrove areas but also misclassifying a large amount of non-mangrove regions. In the two study areas with relatively fewer mangrove samples, the performance of the FCN model is further degraded. This is because, with fewer samples,

FCN is more prone to interference from surrounding similar spectra. It becomes overly sensitive to the mangrove spectral features, leading to the misclassification of many non-mangrove objects with similar spectra to mangroves. In the Shankou Nature Reserve, the DNN model exhibits low recall, F1 score, and IoU, indicating significant issues with missed detection of mangroves in this area. Additionally, the overlap between the predicted results and the true mangrove regions is low, resulting in poor overall performance.

As illustrated in Fig. 7, the segmentation results of each algorithm in the mangrove detection task were visualized to facilitate performance comparison across different study areas. Figure 7 displays the segmentation results of the multispectral image data for the FragMangro, SVM, RF, KNN, FCN, and DNN models. In these visualizations, black regions denote areas identified as mangrove canopies, while white regions represent non-mangrove areas. In the four study areas, there are significant differences in the performance of each algorithm. FragMangro demonstrates higher accuracy and coherence in mangrove identification, while SVM, KNN, RF, FCN and DNN show varying degrees of misclassification and omission.

In the Beilun Estuary Reserve, the results of SVM, KNN, and DNN exhibit significant noise, often misclassifying non-mangrove areas as mangroves, resulting in poor detection performance. The performance of FCN is relatively the worst, as it frequently misidentifies water bodies as mangroves. In contrast, both FragMangro and RF achieve relatively accurate mangrove identification. However, RF shows sparse detection results in smaller mangrove patches (outlined in

Table 3 Comparison table of model evaluation metrics

Region	Mangrove Canopy	Other	Model	Accuracy	Precision	Recall	F1 Score	IoU
Beilun Estuary	592	20313	SVM	97.92%	83.78%	59.19%	69.37%	53.10%
			RF	98.51%	69.59%	75.46%	72.41%	56.75%
			KNN	98.19%	84.97%	63.35%	72.58%	56.96%
			FCN	69.06%	8.17%	97.64%	15.08%	8.16%
			DNN	97.57%	62.76%	76.86%	69.10%	52.78%
			FragMangro	98.88%	95.44%	84.71%	89.75%	81.41%
Shankou	885	31447	SVM	97.29%	75.93%	50.34%	60.54%	43.41%
			RF	98.56%	64.86%	78.95%	71.22%	55.30%
			KNN	98.45%	75.25%	70.25%	72.67%	57.07%
			FCN	95.32%	36.00%	91.78%	51.71%	34.87%
			DNN	97.88%	82.57%	28.27%	42.11%	26.67%
			FragMangro	98.82%	73.76%	81.47%	77.42%	63.16%
Maowei Sea Area 1	919	7085	SVM	95.19%	94.67%	72.14%	81.88%	69.32%
			RF	95.63%	79.54%	81.86%	80.68%	67.62%
			KNN	95.13%	87.05%	74.70%	80.40%	67.23%
			FCN	91.65%	57.71%	96.95%	72.35%	56.68%
			DNN	93.82%	67.36%	88.03%	76.32%	61.71%
			FragMangro	97.07%	87.70%	86.20%	86.95%	76.91%
Maowei Sea Area 2	131	3300	SVM	96.74%	92.37%	54.26%	68.36%	51.93%
			RF	97.32%	87.02%	60.32%	71.25%	55.34%
			KNN	95.98%	95.42%	48.64%	64.43%	47.53%
			FCN	77.45%	13.99%	97.71%	24.47%	13.94%
			DNN	95.69%	46.12%	90.84%	61.18%	44.07%
			FragMangro	98.83%	83.21%	85.16%	84.17%	72.67%

red), failing to fully capture the canopy in these areas. FragMangro performs better in detecting small mangroves within the red-circled areas.

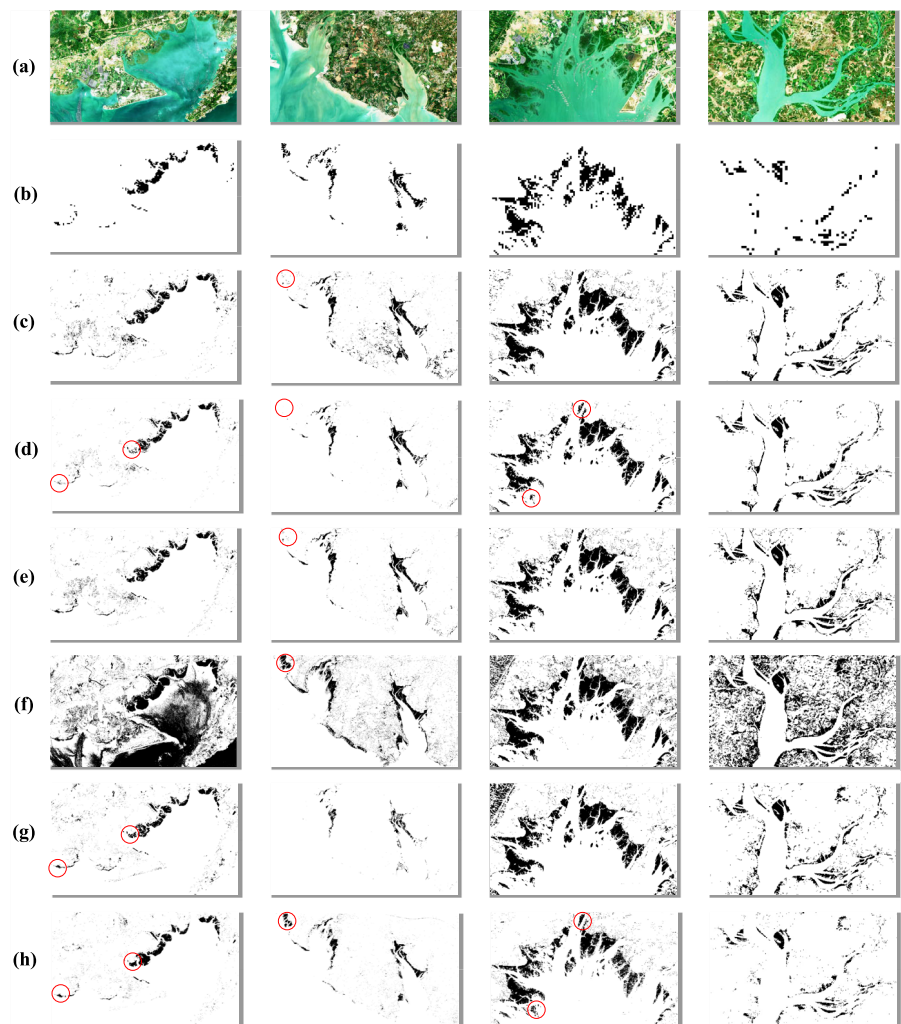
In the Shankou Nature Reserve, the FragMangro model demonstrated significantly higher mangrove identification accuracy compared to other models. The model was able to accurately identify a small mangrove patch in the upper-left corner (marked by a red circle), with clear and coherent boundaries. However, other models showed suboptimal performance in the same area. For instance, the RF model exhibited significant omission errors, while the SVM and KNN models detected only a portion of the mangrove patches and introduced a considerable number of false positives in non-mangrove areas. FCN is able to identify the small mangrove patch in the top-left corner, but there are numerous instances of misclassification throughout the region. In contrast, the DNN model suffers from significant missed detections in this study area. It is worth noting that the Shankou Nature Reserve demonstrated relatively low IoU values compared to the other four study areas. This could be attributed to the complex background information in this

region. The mangroves in the Shankou Nature Reserve are relatively scattered, primarily concentrated at both ends of the peninsula. Notably, the mangroves in the western region are sparse, surrounded by large mudflat areas, which are prone to interference from mixed pixels. This complex environment has led to a certain degree of misclassification by the FragMangro model in this region. However, compared to other algorithms, FragMangro still demonstrated superior predictive performance.

A similar situation was observed in Study Area 2 of the Maowei Sea Reserve. The mangrove patches in this area are relatively small and distributed along both sides of the Dafeng River, exhibiting strong spatial dispersion. In this context, the FragMangro model also demonstrated superior identification performance compared to other algorithms, further validating its robustness and applicability in complex environments.

In the Maowei Sea Reserve Area 1, mangroves occupy a relatively large proportion of the area, but smaller mangrove patches are also distributed in the surrounding regions. In terms of identification performance in this area, the FragMangro model and the RF model performed comparably, both

Fig. 7 Comparison of visualization results. (a) Original image; (b) Label image; (c) SVM; (d) RF; (e) KNN; (f) FCN; (g) DNN; (h) FragMangro



effectively identifying large mangrove regions and demonstrating good boundary continuity. However, the SVM, KNN, FCN and DNN models exhibited more frequent misclassification, particularly introducing more false positives in non-mangrove areas, which led to relatively poorer overall identification results. Additionally, a vegetation area in the upper-left corner of this region, with spectral characteristics similar to mangroves, posed a challenge for the identification task. All four models showed a certain degree of misclassification in this area. Nevertheless, the FragMangro model produced fewer misclassifications, further demonstrating its superior robustness in complex background environments compared to other models.

The analysis shows that the FragMangro model effectively captures the complex spatial structures and spectral features of mangroves through multi-layer convolution operations, extracting multi-level features. In contrast, SVM, KNN, and RF models have limited feature extraction capabilities, struggling to fully utilize the inherent high-dimensional features of multispectral remote sensing data, which leads to more

frequent misclassification and omission. Both FCN and DNN exhibit poor extraction performance due to the small sample size and the influence of complex background spectral information. The FragMangro model achieves higher accuracy, more precise boundaries, and better coherence in both large continuous mangrove areas and small dispersed patches.

6.2 The impact of different convolutional kernel combinations on the detection accuracy of the FragMangro model

The FragMangro model incorporates a combination of 1x1, 1x3, and 1x5 convolutional kernels to construct a multi-scale feature extraction module. To evaluate the impact of this multi-scale convolutional kernel combination on model performance, we conducted three sets of ablation experiments. The results of these experiments are presented in Table 4:

As presented in Table 4, 1, 3, 5, and 7 correspond to convolutional kernels of sizes 1×1, 1×3, 1×5, and

Table 4 Comparison table of experimental results with different convolutional kernels

Region	Conv Group	Accuracy	Precision	Recall	F1 Score	IoU
Beilun Estuary	1	94.88%	85.93%	36.30%	51.04%	34.27%
	1 3	98.92%	85.78%	80.72%	83.17%	71.19%
	1 3 5	98.88%	95.44%	84.71%	89.75%	81.41%
	1 3 5 7	98.85%	79.97%	82.49%	81.21%	68.37%
Shankou	1	97.80%	45.96%	65.15%	53.90%	36.89%
	1 3	97.50%	43.74%	56.92%	49.47%	32.86%
	1 3 5	98.82%	73.76%	81.47%	77.42%	63.16%
	1 3 5 7	98.09%	44.08%	77.89%	56.29%	39.17%
Maowei Sea Area1	1	95.42%	76.15%	73.02%	74.56%	59.43%
	1 3	95.05%	70.66%	72.49%	71.56%	55.72%
	1 3 5	97.02%	87.70%	86.20%	86.95%	76.91%
	1 3 5 7	93.88%	82.42%	61.37%	70.36%	54.27%
Maowei Sea Area2	1	98.51%	52.91%	65.19%	58.42%	41.26%
	1 3	97.94%	60.54%	48.39%	53.78%	36.78%
	1 3 5	98.83%	83.21%	85.16%	84.17%	72.67%
	1 3 5 7	98.24%	70.40%	54.33%	61.33%	44.23%

1×7 , respectively. A comparison of the experimental results clearly reveals the impact of different convolutional kernel combination on model performance. The model employing only the 1×3 convolutional kernel exhibits poor performance across all test areas. For instance, in the Beilun Estuary, the recall is merely 36.30%, with an IoU of 34.27% and an F1 Score of 51.04%, representing the lowest performance among the four experiments. This suggests that using a single-scale convolutional kernel is insufficient to capture the multi-scale features of mangroves effectively, resulting in a significant decline in model performance.

When a combination of 1×1 and 1×3 convolutional kernels is utilized, the model's performance improves in the Beilun Estuary and Dafengjiang areas, though it still falls short of the original multi-scale combination (1×1 , 1×3 , and 1×5). Furthermore, the prediction performance deteriorates in the other two areas. For example, in Maowei Sea Area 1, the F1 Score drops to 71.56%, significantly lower than the 86.95% achieved by the original multi-scale combination. While the inclusion of the 1×1 convolutional kernel offers advantages in retaining spatial information and reducing computational complexity, the absence of the 1×5 kernel-critical for global feature extraction-impairs the model's ability to capture elongated mangrove structures effectively.

In the third experiment, the addition of a 1×7 convolutional kernel results in slight performance improvements across various metrics; however, the overall performance still does not surpass that of the original multi-scale kernel combination. For example, in the Shankou Mangrove Reserve, the F1 Score remains relatively low at 56.29%. Although the

1×7 convolutional kernel is proficient at extracting larger-scale features, its high computational complexity leads to slower inference speeds, limiting its practical application.

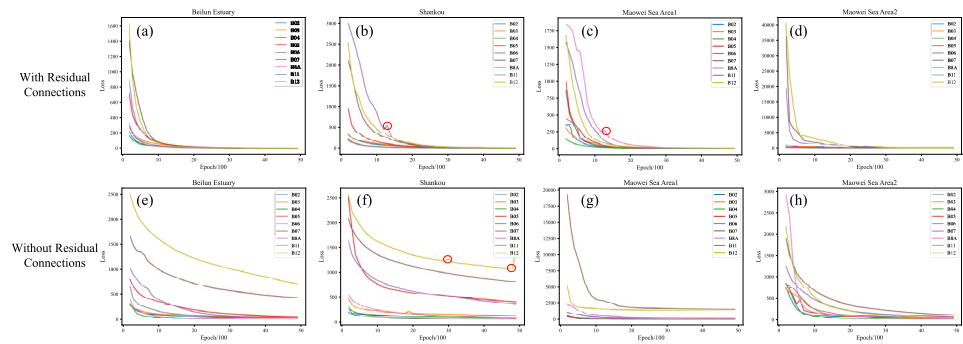
In contrast, the original 1×1 , 1×3 , and 1×5 convolutional kernel combinations consistently outperform other configurations across all evaluation metrics. In all four study areas, FragMangro achieves accuracy, recall, and F1 Score exceeding 70%, with IoU values over 60%. These results demonstrate that the convolutional kernel combinations used by FragMangro are more effective in capturing both fine and global features of mangroves. This approach enhances the identification of dispersed mangroves while maintaining high recognition performance in large mangrove areas.

6.3 The impact of residual connections on the convergence of the FragMangro model

FragMangro incorporates a residual connection module to fuse features across different levels. To assess the impact of these residual connections, an ablation experiment was conducted by removing them. The results of this experiment are presented in Fig. 8.

In Fig. 8, the horizontal axis represents the number of training iterations (Epoch/100), while the vertical axis indicates the loss value (Loss). These curves provide a visual comparison of the convergence behaviour and performance of the two models throughout the training process. The areas where the loss fluctuates sharply are highlighted with red circles in the figure. In all four study areas, the standard

Fig. 8 The chart compares the impact of residual connections on FragMangro loss



FragMangro model exhibits a rapid decrease in loss during the initial training phases, typically stabilizing within the first 3000 epochs. This sharp decline suggests that FragMangro effectively captures meaningful features early in training, achieving a low loss value quickly. Furthermore, the model demonstrates strong stability, avoiding both overfitting and underfitting. The final stabilized loss values consistently remain low, indicating that FragMangro achieves an optimal fit to the training data by the end of the training process. Across all study areas, the loss value stabilizes at a relatively low level after 5000 epochs.

By contrast, the model without residual connections exhibits more pronounced fluctuations during the initial stages of training, with a notably slower rate of loss reduction. This is particularly evident in the Shankou and Beilun Estuary reserves, where the loss varies significantly across different epochs, taking longer to stabilize. This instability suggests a reduced ability of the model to adapt to the training data, making it more susceptible to noise. Although the loss value eventually stabilizes, it remains consistently higher than that of the model with residual connections, indicating a decline in the model's overall fitting capability.

The results highlight the crucial role of residual connections in enhancing the performance of the FragMangro model for intelligent detection of mangrove canopies through remote sensing. The inclusion of residual connections significantly improves both training efficiency and overall model performance by accelerating convergence. Residual connections provide more direct gradient information, allowing the model to reach optimal solutions more quickly while also reducing fluctuations during training and improving stability. Overall, residual connections are essential for boosting the model's performance.

7 Discussion

7.1 Evaluation of A-LrIr effectiveness

This research is grounded in the exploration of association rules between the multispectral features of mangroves.

However, identifying inherent association patterns within mangrove spectra is challenging, as their spectral characteristics closely resemble those of other vegetation, complicating the process of differentiation and making regression tasks particularly difficult. The successful convergence of FragMangro can be largely attributed to the dynamic strategy employed by the A-LrIr mechanism.

A-LrIr initiates the training process with a fixed learning rate of 0.001 and an iteration parameter of 2000. After this initial phase, a dynamic adjustment strategy is activated, monitoring the convergence loss every 100 iterations. When the loss value decreases steadily (i.e., five consecutive reductions), the learning rate factor is increased by 1.2 times. If an abnormal increase in the loss is detected (specifically, when the rate of loss descent exceeds three times the previous gradient in a single monitoring instance), the learning rate factor is reduced to 0.5 times its current value. In cases where the loss value fluctuates (indicating that the learning rate does not follow a consistent downward trend), the learning rate factor is reduced to 0.9 times its previous value. Additionally, if a stable convergence trend is observed, A-LrIr dynamically extends the training iterations to facilitate further optimization throughout FragMangro's iterative process.

Figure 9 depicts the relationship between convergence curves and learning rate adjustments across nine FragMangro sub-modules in the Beilun Estuary Mangrove National Nature Reserve. The horizontal axis represents training iterations, while the positive vertical axis (blue curve) indicates the learning rate (Lr) for each training session. The negative vertical axis illustrates the loss value throughout the model's iteration process. To effectively demonstrate the relationship between loss values and dynamic changes in learning rate, both loss and Lr values are appropriately scaled.

During each training session, A-LrIr dynamically adjusts the learning rate to achieve optimal convergence. Typically, larger loss values correspond to higher learning rates for optimization. The blue curve shows an oscillating downward trend as the red curve decreases. Not all regression tasks progress smoothly. In Fig. 9, appropriate learning rate parameters are derived through oscillating transformations, effectively controlling the steady decline of the loss.

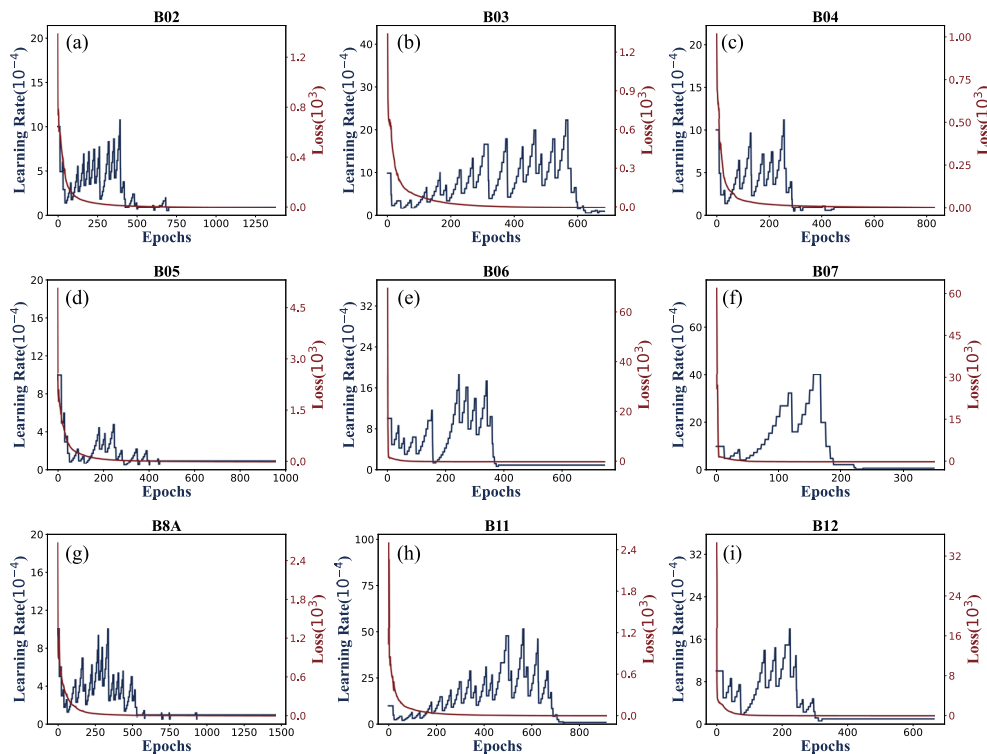


Fig. 9 Convergence curve and learning rate change plot

However, bands B03, B06, B07, B11, and B12 exhibit unstable trends. Although the learning rate mutation mechanism is triggered during training, causing significant oscillations in the learning rate value, A-LrIr re-stabilizes the regression through dynamic parameter adjustments. In the B06, B07, and B12 bands, where initial model loss is relatively high, A-LrIr employs smaller learning rates to control convergence speed. This approach prevents FragMangro from decreasing too rapidly or falling into local optima, ensuring stable regression performance.

To further validate the effectiveness of A-LrIr, this paper compares the loss values obtained using A-LrIr, Adam, and Adagrad after training for the same number of epochs on nine-band datasets from four study areas. Table 5 presents the loss values for these three optimizers across the four study areas.

From the comparison results in Table 5, it can be observed that under the same number of training epochs, the Adagrad algorithm exhibits relatively poorer convergence, with the model converging more slowly. In the Shankou Reserve and Maowei Sea Reserve 1, the Adam algorithm demonstrates better convergence in certain bands. These two reserves, Shankou and Maowei Sea 1, have relatively larger areas and more sample data, making the Adam algorithm more effective in training with large datasets compared to smaller samples. Overall, A-LrIr shows superior convergence compared to Adagrad and Adam. In study areas with fewer data,

such as Maowei Sea Reserve 2, the loss values for some bands have already dropped below 10, indicating a faster convergence rate.

From the table, it can be observed that all three algorithms have not fully converged at the current epoch. The advantage of A-LrIr lies in its ability to automatically increase the number of training epochs, thereby achieving better convergence for each band during training. For ordinary models, suitable convergence is generally reached within 200–800 epochs, but due to the residual stacking design of FragMangro, convergence cannot be achieved in a short period. The number of training epochs required for suitable convergence varies across different bands; some converge around 1000 epochs, while others may require over 3000 epochs. Traditional fixed-epoch training struggles to accurately determine the appropriate number of epochs and necessitates repeated tuning. In contrast, A-LrIr's dynamic training epoch design not only identifies suitable convergence but also significantly saves time and computational resources.

7.2 Comparison of loss between the FragMangro and DACN-M models

To systematically evaluate the convergence effectiveness of the FragMangro model, the A-LrIr strategy was applied to the DACN-M model for comparative analysis. Compared to FragMangro, DACN-M does not incorporate multi-scale

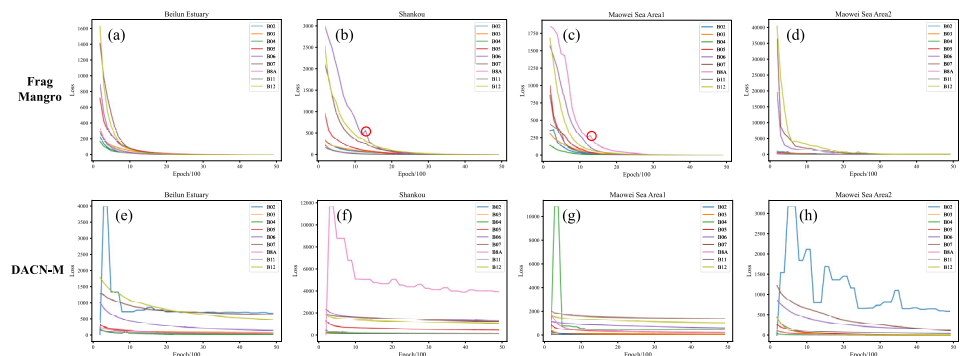
Table 5 Comparison table of loss values for different optimizers under the same round

	Optimizer	Loss								
		0	1	2	3	4	5	6	7	8
Beilun Estuary	Adagrad	553.8	690.8	713.8	2035.4	2235.2	2122.6	897.9	608.0	7212.5
	Adam	258.3	217.4	142.8	986.5	1906.2	641.9	598.9	447.6	1994.7
	A-LrIr	323.8	202.2	132.0	668.5	641.9	383.7	462.7	252.3	871.4
Shankou	Adagrad	706.1	519.8	544.4	2404.9	5750.8	6373.8	2807.5	1035.2	2474.3
	Adam	519.9	259.1	199.5	967.9	2216.9	2760.7	799.2	332.9	1872.8
	A-LrIr	383.7	257.5	297.5	1755.0	1148.4	1793.3	729.0	324.2	1366.0
Maowei Sea Area1	Adagrad	420.0	1078.9	807.1	1965.3	3311.1	4918.2	2954.9	410.8	6032.8
	Adam	493.0	316.0	217.1	581.2	1358.7	1407.1	558.9	178.2	1789.0
	A-LrIr	253.0	255.5	165.7	663.5	364.7	1173.6	1092.0	432.4	1579.1
Maowei Sea Area2	Adagrad	2049.3	3534.0	138.3	1070.7	4322.8	2989.4	2924.8	331.3	8311.8
	Adam	484.1	376.8	94.3	515.6	608.9	181.8	1549.8	111.1	1306.8
	A-LrIr	47.3	291.2	60.7	360.7	11.8	10.0	606.4	109.8	5.9

convolution modules or residual connection modules. Instead, it introduces a channel attention mechanism only after the feature dimension is raised to 128 in the feature dimension expansion module, aiming to further enhance the expressiveness of the features. Apart from this, the remaining design of DACN-M is largely consistent with that of FragMangro. The loss curves of both models, employing the cross-entropy loss function, were examined and compared.

Figure 10 illustrates the loss curve graphs, where a steeper and smoother descent indicates superior model performance. Similar to Fig. 8, the horizontal axis represents the number of training epochs (Epoch/100), while the vertical axis represents the loss value (Loss). The red circles in the figure indicate the locations of sudden loss fluctuations. In all four study areas, the FragMangro model's loss values for all bands decrease rapidly during the initial iteration phase, stabilizing within 3000 epochs. Conversely, DACN-M exhibits greater volatility in some bands, with higher post-stabilization loss values. In certain instances, the DACN-M model fails to achieve full convergence even after 5000 epochs.

In the Beilun Estuary and Maowei Sea Reserve Area 2, the FragMangro model demonstrates stability, converging to lower loss values within 2000 epochs. This indicates effective adaptation to the multispectral data of these regions and rapid extraction of relevant mangrove features. In Maowei Sea Reserve Area 2, the FragMangro model initially exhibits higher loss values, particularly in bands B07 and B12, with initial losses reaching tens of thousands. However, as training progresses, loss values decrease rapidly, stabilizing around the 1000 epochs. Despite the eventual convergence in Maowei Sea Area 2, the initial losses are significantly higher compared to other regions, attributable to more dispersed and relatively smaller mangrove areas, along with data complexity. Nevertheless, the FragMangro model achieves satisfactory convergence after sufficient iterations. The DACN-M model, in contrast, shows instability in both regions. Notably, the B02 band experiences significant loss fluctuations during the first 1000 epochs, indicating challenges in feature extraction. While the DACN-M model eventually improves, with loss values stabilizing around the

Fig. 10 The chart compares the loss of FragMangro and DACN-M

2000 epochs-especially in bands B03, B04, and B05-the post-convergence loss values remain higher than those of the FragMangro model, with more pronounced initial fluctuations.

In the Shankou Reserve and Maowei Sea Reserve Area 1, the FragMangro model continues to exhibit good convergence, with only minor fluctuations in the B07 and B8A bands. The loss values in these bands quickly converge to very low levels. Conversely, the DACN-M model's loss curves are more volatile, particularly in the B04 and B8A bands. In the Shankou Reserve, the B8A loss values oscillate throughout training and do not fully converge within 5000 epochs, indicating the DACN-M model's struggle to effectively extract features from this band. The distinct spectral information in the B8A band may explain the model's difficulty in adaptation. Other bands also show initial fluctuations, though most stabilize after approximately 1000 epochs. The DACN-M model exhibits relatively slower convergence in the Shankou Reserve and greater instability in specific bands compared to the FragMangro model.

The FragMangro model demonstrates rapid and stable convergence across all study areas. Its quick adaptability to data from different regions highlights its strong generalization capability. FragMangro's loss curves are smooth and consistent, with different band combinations nearly overlapping, reflecting high stability in handling multi-band data. This allows it to effectively distinguish between mangroves and surrounding non-mangrove areas. Whether dealing with simple or complex mangrove distributions, the integration of FragMangro's architecture with A-LrIr enables quick adaptation and efficient extraction of key features, achieving precise identification of dispersed mangroves.

7.3 Comparison of detection performance Between the FragMangro and DACN-M models

Figure 11 shows the mangrove detection results of the FragMangro and DACN-M models in the four study areas. Similar to Fig. 7, the black part of the figure represents the identified

mangroves and the white part represents the non-mangroves. The comparative analysis shows that both models show comparable effectiveness in detecting large mangrove areas. However, there is a clear difference in performance when identifying smaller target areas.

In the detection of extensive mangrove regions, particularly within the Maowei Sea Area 1, FragMangro and DACN-M demonstrated accurate identification of large-scale mangrove distributions. Areas characterized by dense, expansive mangrove coverage were effectively detected by both FragMangro and DACN-M, showcasing robust detection capabilities. The detection results exhibit well-defined edges and continuity in large mangrove areas, indicating strong feature extraction and object recognition capabilities for larger targets in both models. Notably, in the Maowei Sea Area 1, both models effectively identified primary mangrove distribution areas, demonstrating stable performance in handling large-scale targets.

The FragMangro model's advantage becomes more pronounced in the detection of smaller, more dispersed target areas. In regions with sparse mangrove distributions, such as the Beilun Estuary Reserve and the Dafengjiang area of the Maowei Sea Reserve, the FragMangro model excels in identifying small mangrove patches along coastlines and estuaries. It accurately detects these small mangrove areas, particularly along the edges of complex terrains, capturing fine details with well-defined boundaries and strong target continuity. This demonstrates a high capacity for detail recognition in complex, small-target regions. In contrast, the DACN-M model exhibits relatively weaker performance in these areas. Specifically, in the Maowei Sea Area 2, the DACN-M model produces sparser results, failing to detect numerous small mangrove patches along riverbanks and showing noticeable omissions. These limitations in feature extraction and multi-scale target handling, particularly in detecting small targets within complex backgrounds, negatively impact the DACN-M model's accuracy and consistency.

The FragMangro model employs a more refined feature extraction network that fully leverages the spectral information from input images, effectively enhancing the capture of

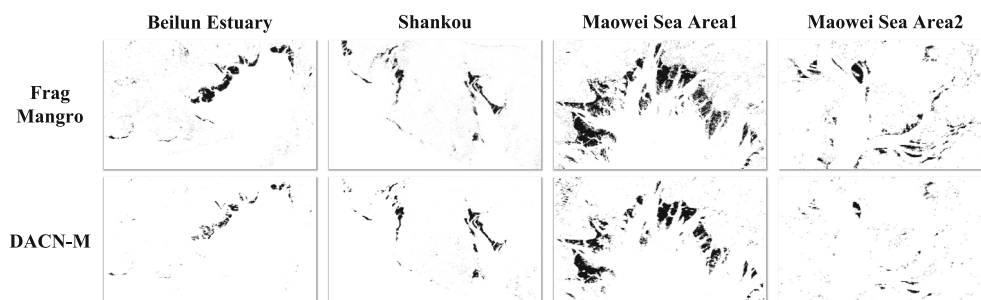


Fig. 11 Comparison chart of detection results: FragMangro vs. DACN-M

details in small target areas. The model is capable of accurately extracting the edges and local features of mangroves, thus enabling precise recognition of small targets. Based on the results of the two ablation experiments presented in Sections 6.2 and 6.3, it is evident that the appropriate application of multi-scale convolution modules, residual connection modules, and channel attention mechanisms is essential for the extraction of dispersed small mangrove patches in complex scenarios. These experimental findings further demonstrate that the specific network structure of FragMangro is more suitable for the extraction of small mangrove patches.

8 Conclusion

To improve the recognition accuracy of dispersed mangroves, we propose the FragMangro model. It demonstrates superior performance in remote sensing-based mangrove monitoring. FragMangro is specifically designed for mangrove detection using Sentinel-2 satellite data. To evaluate its performance, we selected three study areas: the Guangxi Beilun Estuary National Mangrove Reserve, the Guangxi Shankou Mangrove National Nature Reserve, and two areas within the Guangxi Maowei Sea Mangrove Reserve. Comparative analysis of detection results across these regions shows that FragMangro consistently achieves high accuracy. Notably, FragMangro excels at detecting small, dispersed mangrove patches, showing strong adaptability. By integrating multi-scale convolutional kernels and skip connections, the model improves detection accuracy and generalization across different scales and environmental conditions.

A key feature of the FragMangro model is its adaptive learning rate with iterative rounds of dynamic adjustment (A-LrIr). A-LrIr continuously adjusts the learning rate during optimization, improving convergence efficiency and reducing the risk of overfitting. When applied to datasets from multiple study areas, A-LrIr significantly improved the model's performance, leading to better training outcomes. In all study areas, FragMangro outperformed traditional methods, achieving higher accuracy, recall, and F1 Score. These results confirm the model's effectiveness in mangrove detection. Despite its advantages in mangrove remote sensing, FragMangro's robustness needs further validation on larger and more diverse datasets. Additionally, occasional misidentifications still occur in some regions. Future research should expand dataset diversity and incorporate additional remote sensing data sources. These improvements will enhance the model's generalization ability, reduce misidentifications, and improve overall mangrove detection performance.

Data availability statement Data will be made available on request.

Declarations

Competing Interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval This study did not involve human participants or animals, and ethical approval was not required.

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